Problem 1 — Observations

- LeNet trained 15 epochs on MNIST with SGD lr=0.01, batch=64
- val_loss starts rising around epoch 8–10 while train_loss keeps falling → overfitting
- final test accuracy: ~98.5%
- plain SGD converges slowly but reaches decent accuracy on MNIST

Problem 2 — Observations

- Adam converged fastest — adaptive per-parameter learning rates
- SGD+momentum converged faster than plain SGD, smoother updates
- plain SGD slowest — fixed lr, no gradient memory
- Adam reached best final val accuracy overall

Problem 3 — Observations

- lr=0.001, bs=32 gave best results — small stable updates, implicit regularisation
- lr=0.1, bs=128 was worst — large LR + large batch caused divergence
- high LR with large batches = sharp gradients + big steps = unstable training
- smaller batches add noise that helps escape local minima

Problem 4 — Observations

- No Reg: highest train-val gap → most overfitting
- Dropout only: reduced gap, slightly lower train acc
- BatchNorm: faster convergence, moderate gap
- Both (Dropout+BN): lowest gap, best generalisation — winner
- Combining both regularisers gives the most stable training

Problem 5 — Observations

- ReduceLROnPlateau: reactive — drops LR only when val_loss stagnates, step-wise curve
- Cosine Annealing: predetermined smooth decay from max to near-zero

- Cosine gave smoother accuracy curve; Plateau more adaptive to training dynamics
- both outperformed fixed LR on longer training runs